Huawei Access Node KR-ACC-15 Major Optical Transport Failure SOP

Node: KR-ACC-15 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_6166

Description

This Standard Operating Procedure (SOP) outlines the steps to troubleshoot and resolve a Major Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on Huawei Access Node KR-ACC-15, located at KR Puram Access. The Optical Transport Failure is a critical failure domain that affects the Access role of the node, resulting in a Major alarm (ALARM_6166). The failure is caused by a combination of hardware and protocol mechanisms. The SFP/XFP modules, responsible for transmitting and receiving optical signals, may degrade over time due to wear and tear, or suffer from fiber cuts caused by accidents or planned maintenance. The Rx/Tx power loss can be attributed to the degradation of the SFP/XFP modules or the fiber itself. The Optical Transport Failure can be further exacerbated by the use of Wavelength Division Multiplexing (WDM) technology, which multiplexes multiple signals onto a single fiber, increasing the risk of signal degradation and power loss. The Huawei Access Node KR-ACC-15, being a critical component of the network, requires prompt attention to prevent network outages and data loss. This SOP outlines the diagnostic and resolution steps to ensure a timely resolution of the issue.

Pre-requisite Checks

- Verify that the site is accessible and that all necessary safety precautions are in place before proceeding.

- Check the redundant link status to ensure that it is functioning properly.

- Dispatch remote hands to KR Puram Access to assist with the troubleshooting and resolution process.

Potential Root Causes

Cause	Probability
SFP/XFP Module Degradation	High
Fiber Cut	Medium
Rx/Tx Power Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical transceiver , Expected Output: Optical Transceiver Information, Description: This command displays the optical transceiver information, including the SFP/XFP module status, temperature, and optical power. This information helps to identify any issues with the SFP/XFP module or the fiber itself.
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```
Command: display optical transceiver detail, Expected Output: Detailed Optical Transceiver Information, Description: This command displays the detailed optical transceiver information, including the SFP/XFP module temperature, optical power, and error counts. This information helps to identify any issues with the SFP/XFP module or the fiber itself.

Command: display optical transceiver error, Expected Output: Optical Transceiver Error Information, Description: This command displays the optical transceiver error information, including the error counts and error types. This information helps to identify any issues with the SFP/XFP module or the fiber itself.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP Module, Command: replace optical transceiver with , Description: Replace the faulty SFP/XFP module with a new one, ensuring that the new module is compatible with the node and the network configuration. Verify that the new module is securely seated and that all connections are properly made.

Step 2: Step: Repair or Replace the Fiber, Command: repair or replace fiber , Description: Repair or replace the damaged fiber, ensuring that the new fiber is compatible with the node and the network configuration. Verify that the new fiber is securely connected to the SFP/XFP module and that all connections are properly made.

Step 3: Step: Adjust the Rx/Tx Power, Command: adjust optical transceiver power, Description: Adjust the Rx/Tx power of the SFP/XFP module to ensure that it is within the recommended range. Verify that the power adjustment does not affect the network performance or cause any other issues.

Post-Check Verification

```
Command: display optical transceiver , Expected Output: Optical Transceiver Information, Description: Verify that the SFP/XFP module is functioning properly and that the optical power is within the recommended range.

Command: display optical transceiver detail, Expected Output: Detailed Optical Transceiver Information, Description: Verify that the SFP/XFP module temperature, optical power, and error counts are within the recommended range.
```

Roll-back Procedures

- Step: Restore the Original SFP/XFP Module, Command: restore optical transceiver with , Description: Restore the original SFP/XFP module that was replaced, ensuring that the module is securely seated and that all connections are properly made.

- Step: Restore the Original Fiber, Command: restore fiber , Description: Restore the original fiber that was repaired or replaced, ensuring that the fiber is securely connected to the SFP/XFP module and that all connections are properly made.